Assignment 1: Neural Networks
IMDB Movie Review Classification

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Executive Summary

This analysis evaluated multiple neural network configurations for classifying IMDB movie reviews as positive or negative. A baseline neural network was established and then modified to examine how the number of hidden layers, number of hidden units, loss function, activation function, dropout, and L2 regularization affected model performance. Validation accuracy was used to compare the alternative configurations.

Overall, the different configurations produced similar validation results, with peak validation accuracy ranging from approximately 88.6% to 89.0%. The baseline model, consisting of two hidden layers with 16 units per layer, ReLU activation, and binary crossentropy loss, achieved the highest observed validation accuracy of 89.02%. The final model was retrained for four epochs and achieved 88.30% accuracy on the test data, indicating that it correctly classified approximately 88% of previously unseen movie reviews.

Data and Methodology

The analysis used the IMDB movie review dataset, which contains 50,000 reviews labeled as either positive or negative. The dataset is evenly divided into 25,000 reviews for training and 25,000 reviews for testing. For this analysis, the data was limited to the 10,000 most frequently occurring words.

Before training the neural network, each review was converted into a 10,000-dimensional numerical vector indicating which words were present in the review. From the original training data, 10,000 reviews were temporarily separated to create a validation set. The remaining 15,000 reviews were used to train each experimental model. This allowed different model configurations to be compared using data that was not directly used to train them.

The baseline neural network contained two hidden layers with 16 units each and used the ReLU activation function. A sigmoid output layer produced the final positive-or-negative classification. The model used RMSprop as the optimizer and binary crossentropy as the loss function.

A series of experiments then changed individual elements of the baseline model, including the number of hidden layers, number of hidden units, loss function, activation function, and regularization method. Validation performance was compared across these configurations before the selected model was evaluated on the separate test dataset.

Model Experiments and Results

Several neural network configurations were evaluated to determine whether changes to the baseline model could improve validation performance. Each experiment modified a specific aspect of the model while keeping the remaining settings consistent whenever possible. The highest validation accuracy achieved by each configuration was used for comparison.

Model Configuration	Best Validation Accuracy	Best Epoch
Baseline	89.02%	4
1 Hidden Layer	88.96%	4
3 Hidden Layers	88.64%	4
32 Units	88.58%	5
64 Units	88.62%	5
MSE Loss	88.70%	4
Tanh Activation	88.83%	4
Dropout	88.85%	9
L2 Regularization	88.64%	3

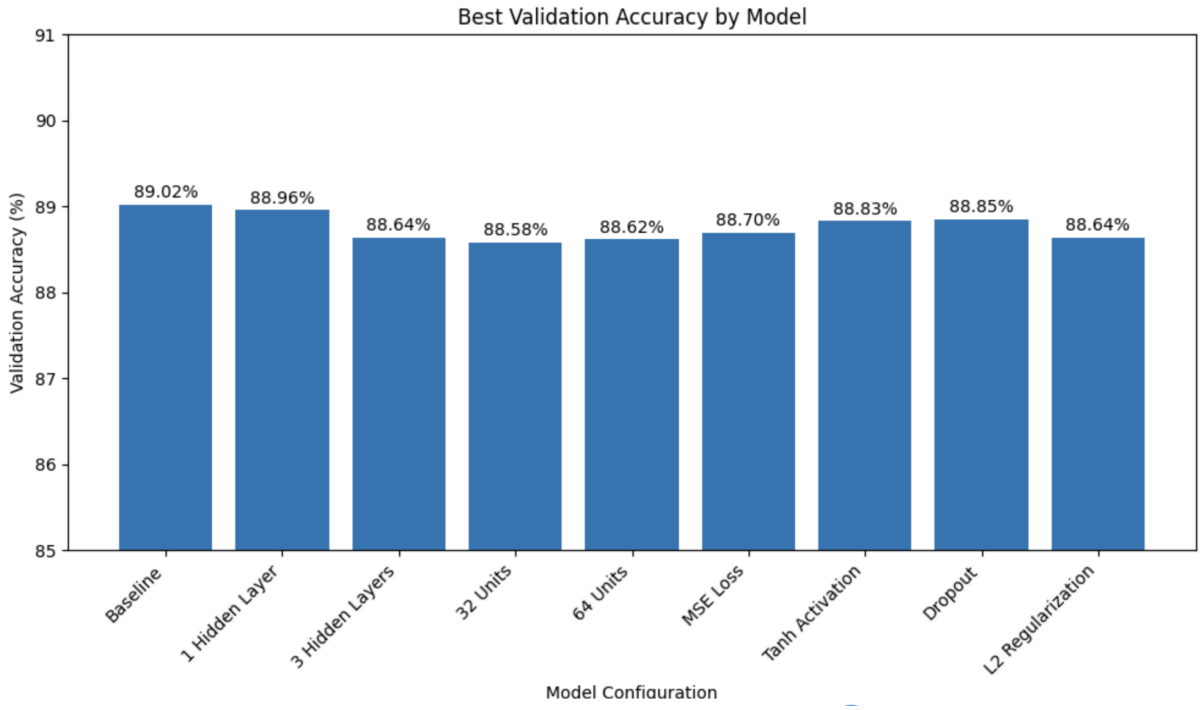


Figure 1. Best validation accuracy achieved by each neural network configuration.

The results show that validation performance remained relatively consistent across all configurations, ranging from 88.58% to 89.02%. The baseline model achieved the highest observed validation accuracy at 89.02%. Increasing the number of hidden layers or hidden units did not improve peak validation performance. Similarly, replacing binary crossentropy with MSE or replacing ReLU with tanh resulted in slightly lower validation accuracy.

The regularization experiments also did not exceed the baseline model's peak validation accuracy. However, dropout reached its best validation accuracy at epoch 9 rather than epoch 4, indicating that it allowed the model to train longer before reaching its best validation performance. Overall, the experiments showed that adding complexity to the baseline model did not produce a meaningful improvement in validation accuracy.

Final Model Performance

Based on the validation results, the baseline configuration was selected for final evaluation. This model contained two hidden layers with 16 units each, used ReLU activation in the hidden layers, binary crossentropy as the loss function, and RMSprop

as the optimizer. Because the model achieved its highest validation accuracy at epoch 4, the selected configuration was retrained for four epochs using the full training dataset before being evaluated on the test dataset.

The final model achieved a test accuracy of 88.30% with a test loss of 0.2915. This means the model correctly classified approximately 88% of the previously unseen movie reviews as positive or negative. The test accuracy was slightly lower than the highest validation accuracy of 89.02%, but the similar results indicate that the model maintained comparable performance when applied to previously unseen data.

Conclusions and Recommendations

The experiments demonstrated that modifying the baseline neural network did not produce a substantial improvement in validation accuracy. Across all configurations tested, the highest validation accuracies remained within a relatively narrow range of 88.58% to 89.02%. The original baseline configuration achieved the highest observed validation accuracy of 89.02%, indicating that increasing model complexity was not necessary to achieve strong performance on this dataset.

Increasing the number of hidden layers or hidden units did not improve validation accuracy, and changing the loss function from binary crossentropy to MSE or the activation function from ReLU to tanh also resulted in slightly lower performance. Dropout and L2 regularization did not increase peak validation accuracy. However, dropout reached its highest validation accuracy at epoch 9 rather than epoch 4, suggesting that it helped delay the model's tendency to overfit.

Based on these results, the baseline configuration is the recommended model among the configurations evaluated in this analysis. It provided the highest observed validation accuracy while maintaining a relatively simple architecture. The final model achieved 88.30% accuracy on the test dataset, demonstrating that the selected configuration maintained similar performance when applied to previously unseen movie reviews.

More broadly, the analysis demonstrates that a more complex neural network does not necessarily produce better results. Model performance should be evaluated using validation data rather than assuming that additional layers, units, or other modifications will automatically improve predictive accuracy.